

Czech Technical University in Prague
FJFI - Department of Physics
02MECH - Mechanics
Test 1

20.11.2024 / 12:00pm

Name					
Problem	1	2	3	4	Total
Score					

Instructions

1. The time allocated for the exam is 90 minutes.
2. The maximum score for the test is 4 points.
3. Each question is worth 1 point, but the questions may not require the same amount of time. Therefore, it is advisable to read the entire assignment before starting to solve it.
4. During the exam, you are not allowed to use any class materials or ask your neighbor for help.
5. Calculators are not allowed. You are not allowed to keep any electronic devices on your desk.
6. Use only pens or non-erasable ink pens for the test; erasable pens are not allowed.
7. Write your answers legibly. Illegible answers will not be scored.
8. Write your answers in the provided space. If you need more space, use the back of the sheet.
9. Two scratch papers will be provided. You may use these papers for trial and error, but do not return them.

Question 1

Two particles are moving uniformly in two-dimensional space on straight lines. The first particle starts at point $\vec{r}_1 = (0, 4)$ m with velocity $\vec{v}_1 = (1, -2)$ m/s, and the second particle starts at point $\vec{r}_2 = (-3, 2)$ m with velocity $\vec{v}_2 = (2, 1)$ m/s. Find:

- (a) The vector of their relative position with respect to time, $\vec{R}(t)$. — *0.5 points*
- (b) The moment t_c of closest approach and its distance. — *0.3 points*
- (c) The intersection point of their trajectories. — *0.2 points*

Solution:

Question 2

An object is thrown vertically downward from a height h with an initial velocity v_0 . The motion takes place on earth with gravitational acceleration g and air resistance is neglected.

- (a) What should be v_0 so that the object lands τ seconds earlier than it would if it was released from rest? — *0.7 points*
- (b) Assume that there was also a force acting in the perpendicular direction as the throw of the object and accelerating it as $a = \frac{g}{\sqrt{3}}$. What would be the range of the object in this case? — *0.3 points*

Solution:

Question 3

A particle of mass m can move without friction along the x -axis. At time $t = 0$, it is at rest at position $x = 0$. A force $F_x(t) = a + bt$ (where a and b are constants) acts on it.

- (a) Determine the acceleration $a(t)$, the velocity $v(t)$, the displacement $x(t)$. — *0.6 points*
- (b) Determine the power delivered by the force at time t . — *0.4 points*

Solution:

Question 4

A simple pendulum with mass m and length l , initially displaced by an angle ϕ_0 from the vertical and given an initial tangential speed v_0 at $t = 0$. Find the oscillation amplitude ϕ_1 .
— 1 point

Solution: